How Criminals Make Decisions

A Microeconomic Introduction to Game Theory

Intermediate Microeconomics · Prof. Naveen Sunder · EC-224 · Mini Lecture Handout

Before You Begin: Quick Self-Check. Can you answer these questions? If not, they will be covered in class — but think about them first.

1. Two suspects are caught by police. Each can confess or stay silent. They cannot communicate. Why might *both* confess even if staying silent is better for both of them combined?
2. What does it mean for one player's strategy to be a "best response" to another player's strategy?
3. Can you think of a real-world situation where two firms (or countries) face the same kind of dilemma as the two suspects?

Questions This Handout Will Answer

- How do economists model situations where your best choice depends on what others do?
- What is a Nash Equilibrium, and how do you find one?
- Why do individually rational choices sometimes produce collectively bad outcomes?
- How does order of play change the analysis?
- Where do these ideas show up in business and technology today?

Notation Used in This Handout

Symbol	Meaning
S_i	Strategy of player i
π_i	Payoff (utility) of player i
BR_i	Best response of player i
NE	Nash Equilibrium
$\succcurlyeq$	"at least as good as"

What Is Game Theory?

Game theory is the branch of economics that studies **strategic behaviour**: situations where the best action for one person depends on what others do.

In standard consumer or firm theory, a decision-maker faces fixed prices and maximises subject to a constraint. In game theory, there are *multiple* decision-makers, and each one's optimal choice depends on the choices of everyone else. This interdependence is what makes a situation a "game."

Definition

A **game** consists of three elements:

1. A set of **players** $\{1, 2, \ldots, n\}$
2. A set of **strategies** S_i available to each player i
3. A **payoff function** $\pi_i(s_1, s_2, \ldots, s_n)$ giving the payoff to player i for every possible combination of strategies

Key Idea

Strategic vs. Non-Strategic Decisions. Choosing how much coffee to buy at a fixed price is a non-strategic decision — the price does not depend on your choice. Choosing how aggressively to price a new product when a competitor is watching your every move *is* strategic. Game theory handles the second kind.

The Prisoner's Dilemma

The Prisoner's Dilemma is the most famous game in economics. It illustrates a deep and uncomfortable truth: individually rational behaviour can produce collectively bad outcomes.

The Classic Setup

Two suspects, Bonnie and Clyde, are arrested and held *separately*. Each can choose to **stay Silent (S)** or **Confess (C)**. The payoffs (in years of prison time, so lower is better) are:

	Clyde: Silent	**Clyde: Confess**
Bonnie: Silent	$(-5,\ -5)$	$(-30,\ -1)$
Bonnie: Confess	$(-1,\ -30)$	$(-10,\ -10)$

Convention: The first number in each cell is Bonnie's payoff; the second is Clyde's.

A Modern Parallel: AI Companies and Regulators

The same mathematical structure arises constantly in business. Consider two AI companies — call them Firm A and Firm B — both caught using each other's stolen proprietary training data. Regulators question them separately. Each can disclose (*confess*) or cover up (*stay silent*).

- If both **cover up**: moderate fines for each.
- If one **discloses and the other covers up**: the discloser gets leniency; the other faces severe penalties.
- If both **disclose**: both face significant penalties, but less than being caught hiding evidence.

The payoff structure is identical to the criminal case. The math is the same; only the context changes. This is what makes game theory so powerful: a single model captures many different real-world situations.

Nash Equilibrium

Definition

A **Nash Equilibrium (NE)** is a set of strategies $(s_1^*, s_2^*, \ldots, s_n^*)$, one for each player, such that *no player can improve their payoff by unilaterally changing their strategy*, given the strategies of all other players.
Formally, for each player i:

$$\pi_i(s_i^*, s_{-i}^*) \;\geq\; \pi_i(s_i, s_{-i}^*) \quad \text{for all } s_i \in S_i$$

where s_{-i}^* denotes the equilibrium strategies of all players *other than* i.

Finding the NE: Best Response Analysis

The standard method for finding a Nash Equilibrium in a two-player game is to identify each player's **best response (BR)** to every possible strategy of the other player.

Step 1. For each of Clyde's strategies, find Bonnie's best response:

- If Clyde plays Silent: Bonnie's payoffs are -5 (Silent) vs. -1 (Confess). Best response: **Confess**.
- If Clyde plays Confess: Bonnie's payoffs are -30 (Silent) vs. -10 (Confess). Best response: **Confess**.

Observation: Confess is Bonnie's best response *regardless* of what Clyde does. This is called a **dominant strategy**.

Step 2. By symmetry, Confess is also Clyde's dominant strategy.

Step 3. The NE is the cell where both players are playing their best responses simultaneously:

$$NE = (\text{Confess},\ \text{Confess}) \quad \Rightarrow \quad \text{Payoff: } (-10,\ -10)$$

Key Idea

The Paradox. The NE gives both players -10 years each. But if both had stayed silent, they would each get -5 years. The outcome $(-5, -5)$ is better for *both* — yet it is not an equilibrium. Each player individually has an incentive to deviate from it. This is the heart of the Prisoner's Dilemma: **individual rationality and collective welfare conflict.**

Dominant Strategy Equilibrium

Definition

A **dominant strategy** for player i is a strategy that gives the highest payoff *regardless* of what any other player does:

$$\pi_i(s_i^*, s_{-i}) \;\geq\; \pi_i(s_i, s_{-i}) \quad \text{for all } s_i \in S_i \text{ and all } s_{-i}$$

When every player has a dominant strategy, the outcome is a **dominant strategy equilibrium** — a special (and particularly compelling) type of NE.

Common Mistake

NE $\neq$ the best joint outcome. Students often confuse the Nash Equilibrium with the outcome that maximises total (combined) payoffs. In the Prisoner's Dilemma, the NE $(-10, -10)$ is *not* jointly optimal — $(-5, -5)$ is. The NE describes what *rational, self-interested* players will do, not what is socially optimal.

Simultaneous vs. Sequential Games

All the analysis so far assumed both players choose **simultaneously** — neither observes the other's action before choosing. Many real-world situations are different: one player moves first, and the other observes and responds.

Definition

- A **simultaneous game** is one in which players choose their strategies at the same time, without observing others' choices.

- A **sequential game** is one in which at least one player observes another's action before choosing their own.

Sequential games are represented using a **game tree** (extensive form) rather than a matrix.

Backward Induction

Definition

Backward induction is the method for finding the equilibrium of a sequential game. Start from the *last* decision node and work backwards to the beginning.

Example. Player A moves first (Top or Bottom); Player B observes and responds (Left or Right). Payoffs are (A, B):

	B: Left	**B: Right**
A: Top	(3, 9)	(1, 8)
A: Bottom	(0, 0)	(2, 1)

Solving by backward induction:

Step 1: What will B do?

- If A played Top: B chooses Left ($9 > 8$).
- If A played Bottom: B chooses Right ($1 > 0$).

Step 2: What will A do? A anticipates B's responses:

- Top $\Rightarrow$ B plays Left $\Rightarrow$ A gets 3.
- Bottom $\Rightarrow$ B plays Right $\Rightarrow$ A gets 2.

A prefers Top ($3 > 2$).

Equilibrium: A plays Top; B plays Left. Outcome: $(3, 9)$.

Key Idea

First-Mover Advantage. In the example above, going first is beneficial for A: by committing to Top, A induces the favourable response from B. This "first-mover advantage" is the intuition behind the **Stackelberg model** of firm competition — a topic covered fully in the main course.

Real-World Applications

Game theory is not an abstract tool. It shapes real decisions in business, policy, and technology.

- **AI development races (OpenAI vs. Google vs. Meta).** Each firm faces a Prisoner's Dilemma: rush a product to market now ("confess" in terms of releasing prematurely) or wait to develop it safely. If all firms wait, the outcome is better for everyone — but individually, there is always an incentive to move first.
- **Salary negotiations.** You and your potential employer both have private information about your outside options. Game theory models the strategic disclosure of this information.
- **US Radio Spectrum Auction.** The US Federal Communications Commission used game theory to design a $60 billion auction for wireless spectrum licences. Economists designed the rules of the game so that bidders would reveal their true valuations.
- **Climate agreements.** Countries deciding whether to reduce carbon emissions face a global Prisoner's Dilemma — each country benefits if others reduce emissions, but each individually prefers not to bear the cost. This is why international agreements are hard to sustain.
- **Antitrust and App Store pricing.** Apple's App Store rules force Spotify to pay a 30% commission. Spotify cannot move to a rival platform cheaply. This is a sequential game where Apple moved first (by building the dominant platform), creating a structural advantage.

A Critical Assumption: Are Players Rational?

Nash Equilibrium requires players to be **rational**: they always choose the action that maximises their own payoff. This is a strong assumption. In practice, humans often deviate from fully rational behaviour due to cognitive biases, social norms, emotions, and incomplete information. **Behavioural economics** — one of the fastest-growing areas of the discipline — studies exactly these deviations and has produced major insights in policy design, marketing, and finance.

Key Idea

Preview. One of the most-cited findings in behavioural economics concerns the **Ultimatum Game**: Player 1 is given \$10 and offers Player 2 some amount x. If Player 2 accepts, both keep their shares; if Player 2 rejects, both get \$0. Rational theory predicts Player 2 should accept any $x > 0$. In experiments, Player 2s routinely *reject* "unfair" offers — even at personal cost. This is not rationality; it is a preference for *fairness*. Behavioural economics incorporates these preferences into economic models.

Chapter at a Glance

Concept	One-Line Summary
Game	Players + Strategies + Payoffs
Strategic behaviour	Your best choice depends on what others do
Payoff matrix	A grid representing all possible outcomes
Dominant strategy	Best response regardless of what others do
Nash Equilibrium	No player wants to deviate, given others' choices
Prisoner's Dilemma	Individual rationality $\neq$ collective optimality
Sequential game	One player observes before the other acts
Backward induction	Solve from the last node backwards
First-mover advantage	Committing early can shape others' responses
Rationality	Players maximise their own payoffs (key assumption)

Key Idea

The thread connecting it all. Game theory takes the tools of optimisation (maximise your payoff) and adds *strategic interdependence* (your payoff depends on others' choices). Nash Equilibrium is the solution concept that ties it together. The Prisoner's Dilemma is the canonical example of why this matters: sometimes the individually rational outcome is collectively terrible — and understanding why is the first step toward designing better outcomes.

Practice Questions

Questions are arranged in three levels. ⋆ easy ⋆⋆ intermediate ⋆⋆⋆ challenging.
Full solutions follow. Attempt every question independently first.

Level 1: Conceptual and Foundational

⋆ **[Reading a Payoff Matrix.]** Consider the following game between Firms X and Y, where the strategies are Price High (H) or Price Low (L), and payoffs represent monthly profits in thousands of dollars:

	Firm Y: High	**Firm Y: Low**
Firm X: High	(10, 10)	(2, 14)
Firm X: Low	(14, 2)	(5, 5)

[(a)]What is Firm X's best response if Firm Y prices High? What is Firm X's best response if Firm Y prices Low? Does Firm X have a dominant strategy? Explain. What is the Nash Equilibrium of this game? Is this a Prisoner's Dilemma? Why or why not?

⋆ **[Identifying a Nash Equilibrium.]** For the game below, identify all Nash Equilibria. If there are multiple, list all of them.

	B: Left	**B: Right**
A: Top	(3, 9)	(2, 8)
A: Bottom	(0, 2)	(1, 1)

⋆ **[Concepts.]** Answer in one or two sentences each.

[(a)]What is the difference between a *dominant strategy* and a Nash Equilibrium? Can you have a NE without dominant strategies? Why does the Prisoner's Dilemma produce a collectively sub-optimal outcome? What does it mean for a player to "play a best response"? **Explain to a friend (no jargon):** In plain English, explain what a Nash Equilibrium is. Your explanation should convince someone who has never studied economics.

Level 2: Intermediate Analysis

⋆⋆ **[AI Companies: Prisoner's Dilemma.]** Two AI companies (Firm A and Firm B) are each deciding independently whether to release a powerful new model **Now (N)** or **Wait (W)** for further safety testing. The payoffs (in market share units) are:

	Firm B: Wait	**Firm B: Now**
Firm A: Wait	(8, 8)	(2, 12)
Firm A: Now	(12, 2)	(4, 4)

[(a)]Find the Nash Equilibrium. Is this a Prisoner's Dilemma? Identify the "socially optimal" outcome and explain why it is not an equilibrium. What real-world mechanism might allow the firms to achieve the socially optimal outcome? (Think: regulation, contracts, reputation.) Now suppose Firm A moves first and Firm B observes before deciding. Draw the game tree and solve by backward induction.

⋆⋆ **[Multiple Nash Equilibria.]** Consider the following coordination game (two firms choosing which technology standard to adopt):

	Firm B: Tech 1	**Firm B: Tech 2**
Firm A: Tech 1	(5, 5)	(0, 0)
Firm A: Tech 2	(0, 0)	(3, 3)

[(a)]Find all Nash Equilibria. Does either firm have a dominant strategy? Explain. Which NE is "better" in terms of total payoffs? Does game theory alone tell you which NE will be reached? **Real-world connection:** Can you think of a technology standard battle from recent history that resembles this game?

⋆⋆ **[Sequential Game: Salary Negotiation.]** A job candidate (Player A) moves first, choosing to ask for a **High** or **Low** salary. The firm (Player B) observes and decides to **Accept** or **Reject**. Payoffs (candidate, firm):

	Firm: Accept	**Firm: Reject**
Candidate: High	(10, 3)	(0, 7)
Candidate: Low	(5, 8)	(0, 7)

[(a)]Solve this game by backward induction. Draw the game tree. What does the candidate choose? What does the firm do? Interpret the result: does the candidate get the outcome they want? Why or why not? How would the outcome change if the firm could *commit* in advance to always accepting High salary requests? Is this credible?

Level 3: Advanced Problems

⋆⋆⋆ **[Climate Negotiations.]** Two countries (Country A and Country B) each decide whether to **Reduce** (R) or **Ignore** (I) carbon emissions. Payoffs represent welfare units:

		B: Reduce	**B: Ignore**
3.	**A: Reduce**	(6, 6)	(1, 9)
	A: Ignore	(9, 1)	(2, 2)

[(a)]Find the Nash Equilibrium. Is this a Prisoner's Dilemma? Now suppose this game is repeated 100 times. Does repetition change the equilibrium? (*Hint: consider what happens in the last round.*) Suppose the game is repeated *indefinitely* (infinite horizon). Argue informally why cooperation might now be sustainable. **Essay (6–8 sentences):** Explain in well-reasoned prose why the Prisoner's Dilemma structure makes international climate agreements so difficult to sustain, and what economic mechanisms have been proposed to overcome this problem. No bullet points.

⋆⋆⋆ **[Designing a Better Game.]** In the classic Prisoner's Dilemma (Bonnie and Clyde), the Nash Equilibrium produces a worse outcome for both players than mutual cooperation.

[(a)]Propose a *change to the payoff structure* that would make mutual silence (S, S) a Nash Equilibrium. What does this change represent in real-world terms (e.g., a contract, a fine for defection, a reputation mechanism)? Show formally (using the best-response logic from Section 3.1) that your proposed payoffs make (S, S) a NE. **Real-world connection:** Which of the following mechanisms does your proposed change most closely resemble? (i) A legally binding cartel agreement; (ii) a regulator offering leniency for first disclosure; (iii) an industry self-regulation code. Explain your choice.

⋆⋆⋆ **[Explain to a Friend — Level 3.]** A classmate says: "Game theory assumes everyone is selfish and rational — it can't tell us anything useful about the real world where people cooperate and make mistakes."

Write a short, well-reasoned response (5–7 sentences) that engages seriously with this critique. Acknowledge what is valid in the objection and explain what game theory can and cannot do. No jargon — write as if explaining to an intelligent non-economist.

Solutions to Practice Questions

3. Attempt all questions independently before reading solutions.

Solution to Level 1, Question 1 (Payoff Matrix)

(a) If Firm Y prices High: Firm X's payoffs are 10 (High) vs. 14 (Low). Best response: **Low**.

(b) If Firm Y prices Low: Firm X's payoffs are 2 (High) vs. 5 (Low). Best response: **Low**.

(c) Yes. Firm X's best response is Low regardless of what Firm Y does. Low is a dominant strategy for Firm X. By symmetry, Low is also dominant for Firm Y.

(d) Nash Equilibrium: **(Low, Low)** with payoffs $(5, 5)$.

(e) Yes, this is a Prisoner's Dilemma. Both firms playing Low gives $(5, 5)$, but both playing High gives $(10, 10)$ — a better outcome for both. The NE is not Pareto optimal. Individual rationality produces a collectively inferior outcome.

Solution to Level 1, Question 2 (Identifying NE)

Best responses for Player A:

- If B plays Left: A gets 3 (Top) vs. 0 (Bottom) $\Rightarrow$ Top.
- If B plays Right: A gets 2 (Top) vs. 1 (Bottom) $\Rightarrow$ Top.

Top is a dominant strategy for A.

Best responses for Player B:

- If A plays Top: B gets 9 (Left) vs. 8 (Right) $\Rightarrow$ Left.
- If A plays Bottom: B gets 2 (Left) vs. 1 (Right) $\Rightarrow$ Left.

Left is a dominant strategy for B.

Nash Equilibrium: (Top, Left) $= (3, 9)$**.** There is exactly one NE.

Solution to Level 1, Question 3 (Concepts)

(a) A dominant strategy is optimal regardless of others' actions; a NE only requires that no player wants to deviate *given* others' equilibrium strategies. You can have a NE even when no player has a dominant strategy — for example, in coordination games (see Level 2, Q2).

(b) In the Prisoner's Dilemma, each player has a dominant strategy to defect (confess). But when both defect, both are worse off than if both cooperated. The problem is that cooperation is not individually incentive-compatible: each player gains by defecting *unilaterally*, even if mutual cooperation is better.

(c) A best response is the strategy that maximises player i's payoff, *given the specific strategy being played by the other player*. It is a conditional optimum, not an unconditional one.

(d) A strong answer: "A Nash Equilibrium is a situation where everyone is doing the best they can, given what everyone else is doing — and nobody wants to change. It is a stable resting point. If everyone is at a NE and you tell each person what the others are doing, no one wishes they had chosen differently."

Solution to Level 2, Question 1 (AI Companies)

(a) By the same best-response logic as the Prisoner's Dilemma: Now is a dominant strategy for both firms. **NE: (Now, Now)** $= (4, 4)$**.**

(b) Yes, this is a Prisoner's Dilemma. (Wait, Wait) $= (8, 8)$ is the socially optimal outcome — it Pareto-dominates the NE. But it is not an equilibrium: if Firm B waits, Firm A prefers Now $(12 > 8)$.

(c) Potential mechanisms: (i) a government regulation requiring safety certification before release; (ii) a binding industry agreement with financial penalties for early release; (iii) a reputational mechanism if the game is repeated — firms that rush and cause harm suffer long-term damage.

(d) Game tree (Firm A moves first):
If A plays Wait $\Rightarrow$ B chooses Now $(12 > 8) \Rightarrow$ A gets 2.
If A plays Now $\Rightarrow$ B chooses Now $(4 > 2) \Rightarrow$ A gets 4.
A prefers Now $(4 > 2)$. **Backward induction NE: (Now, Now)**.
The sequential structure does not resolve the dilemma here.

Solution to Level 2, Question 2 (Coordination Game)

(a) Check all four cells:

- (Tech 1, Tech 1): A gets 5 from Tech 1 vs. 0 from Tech 2 $\Rightarrow$ no deviation. B gets 5 from Tech 1 vs. 0 from Tech 2 $\Rightarrow$ no deviation. **NE.**
- (Tech 2, Tech 2): A gets 3 vs. 0 $\Rightarrow$ no deviation. B gets 3 vs. 0 $\Rightarrow$ no deviation. **NE.**
- (Tech 1, Tech 2) and (Tech 2, Tech 1): Both involve deviation incentives. Not NE.

Two NE: (Tech 1, Tech 1) and (Tech 2, Tech 2).

(b) No dominant strategy for either firm. The best response depends entirely on what the other firm does.

(c) (Tech 1, Tech 1) gives total payoff 10; (Tech 2, Tech 2) gives 6. Tech 1 is better. But game theory alone does not predict which NE is reached — history, coordination devices, and communication matter. This is the "equilibrium selection problem."

(d) Historical examples: VHS vs. Betamax; Blu-ray vs. HD-DVD; USB-C vs. Lightning connector.

Solution to Level 2, Question 3 (Salary Negotiation)

(a) Game tree: Candidate moves first.
If Candidate asks High: Firm gets 3 (Accept) vs. 7 (Reject) $\Rightarrow$ Reject. Candidate gets 0.
If Candidate asks Low: Firm gets 8 (Accept) vs. 7 (Reject) $\Rightarrow$ Accept. Candidate gets 5.
Candidate compares: High $\Rightarrow$ 0; Low $\Rightarrow$ 5. Candidate asks **Low**.

(b) Candidate asks Low; Firm accepts. Outcome: $(5, 8)$.

(c) The candidate does not get the higher salary (10) because they anticipate rejection. Backward induction forces the candidate to internalise the firm's response. This is the "hold-up problem" in sequential bargaining.

(d) If the firm credibly committed to accepting High requests, the candidate would ask High and both are better off: $(10, 3)$ vs. $(5, 8)$ for the firm means the firm would actually prefer to *not* commit, since it prefers outcome $(5, 8)$. Commitment to accepting High is **not credible** — the firm always prefers to reject High if it can.

Solution to Level 3, Question 1 (Climate Negotiations)

(a) Ignore is a dominant strategy for both countries. **NE: (Ignore, Ignore)** $= (2, 2)$**.** This is a Prisoner's Dilemma: (Reduce, Reduce) $= (6, 6)$ is better for both.

(b) In a finitely repeated game, backward induction applies. In the final round (round 100), the game is identical to the one-shot version — (Ignore, Ignore) is the NE. Knowing this, in round 99 there is no future reward for cooperation — again, (Ignore, Ignore). This logic unravels all the way back to round 1. The NE of the 100-round game is (Ignore, Ignore) in every round.

(c) With an infinite horizon, a strategy like "cooperate unless the other player defected last period" (a *grim trigger* strategy) can sustain cooperation. The threat of permanently losing cooperation is a credible punishment that deters defection — provided players value the future sufficiently (discount rate is not too high).

(d) A strong essay answer notes: (i) the payoff structure of international climate negotiations is a Prisoner's Dilemma: each country benefits if others reduce emissions but prefers to free-ride on others' reductions; (ii) without binding enforcement, no country has a unilateral incentive to reduce; (iii) proposed solutions include carbon border adjustments (changing payoffs), multilateral treaties with verification mechanisms (adding repetition and monitoring), and green technology investment funds (changing the payoff structure so cooperation is individually rational).

End of Handout · Intermediate Microeconomics · Prof. Naveen Sunder